

Analyzing Edwards Aquifer Groundwater Quality: Sucralose in Spring Systems

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Introduction

The Edwards Aquifer is one of the world's most productive artesian aquifer systems. It provides water for the agricultural, industrial, and domestic needs of over two-million south-central Texans. The limestone aquifer covers nearly 4,000 square miles, of which roughly 1,250 are exposed at the surface forming the aquifer's recharge zone. South of the recharge zone is the artesian zone, where pressurized water within the confined layers of the aquifer discharge to the surface forming Comal and San Marcos springs. These two spring systems represent the largest artesian springs in the southwest U.S. and their discharge is recorded by the USGS.

In 2013, the U.S. Fish and Wildlife service approved the Edwards Aquifer Authority Habitat Conservation Plan (EAHCP) which detailed the framework for protecting the seven Threatened and Endangered species located in the Edwards Aquifer and its associated springs. The plan includes actions to protect springflow and species aquatic habitats during periods of severe drought.

In addition to flow conditions, water quality has a considerable impact on the long-term well-being of the aquifers' endangered species. The EAHCP contains water quality goals and includes a robust water quality monitoring program as part of its implementation. Part of the water quality monitoring program is aimed at identifying anthropogenic contributions to the aquifer. A particularly well-suited groundwater tracer of wastewater contributions is sucralose (SUC), an artificial sweetener approved for consumption by the FDA in 1998. The large percentage of SUC not metabolized by the human body is excreted into sewage wastewater, and further resists degradation by normal wastewater treatment processes. It is ubiquitously detected in wastewater effluent and environmentally persistent.

In the present study, we examine the concentrations of sucralose collected at Comal and San Marcos springs from the period of 2021– 2025.



Fountain Darter



Comal Springs Riffle Beetle



Texas Wild-Rice

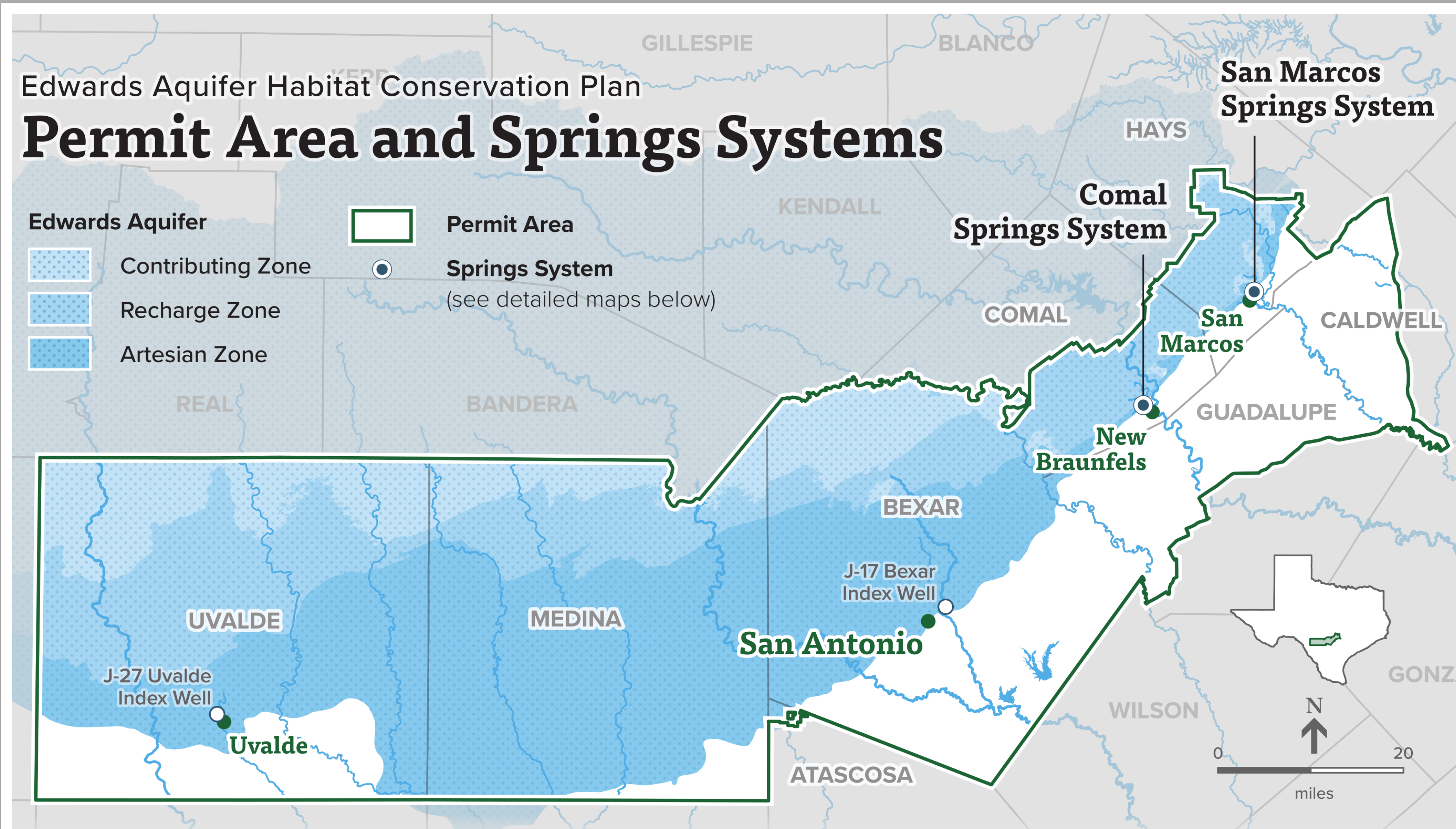


Figure 1: The EAHCP permit area spans portions of eight counties (Uvalde, Medina, Atascosa, Bexar, Comal, Guadalupe, Hays, and Caldwell) and includes areas of the Contributing, Recharge, and Artesian Zones of the Edwards Aquifer. Monthly sucralose sampling is performed at Spring Run 3 which feeds into Landa Lake as part of the Comal system and Hotel Spring which feeds into Spring Lake as part of the San Marcos system (green arrows point to spring locations). These spring complexes form the headwaters of the Comal River and San Marcos River.

Results

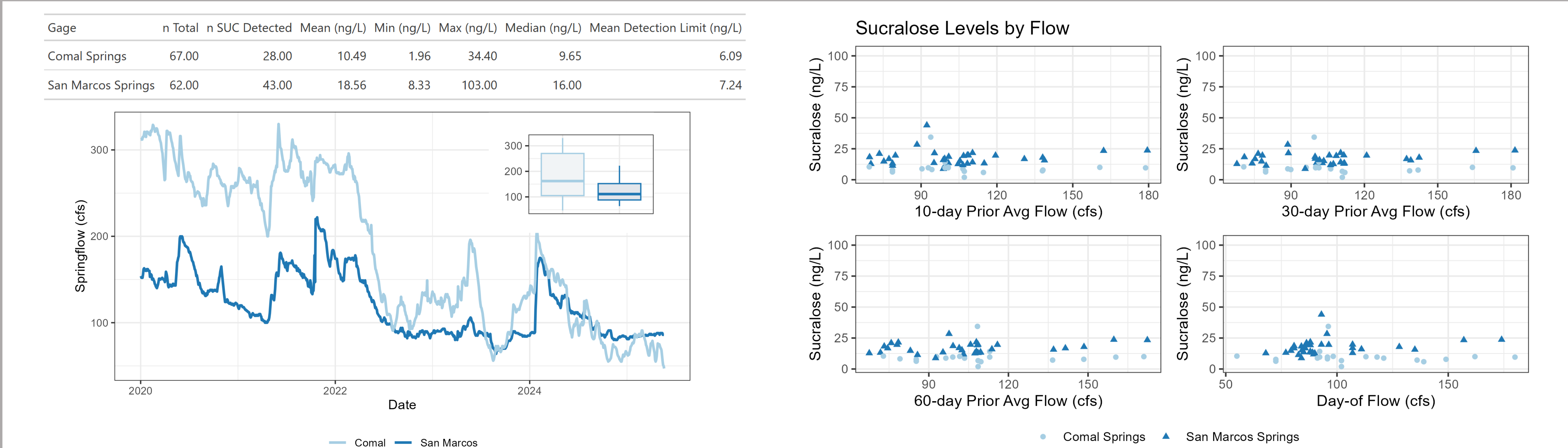


Figure 2: Summary statistics of sucralose concentrations within the Comal and San Marcos springs (top left). Hydrograph (bottom left) detailing daily flow averages from Jan. 2020 to Jun. 2025 for streamgages 08170000 and 08168710 (San Marcos and Comal). Sucralose concentrations in relation to average springflow of 0, 10, 30, and 60 days prior to collection for sucralose testing (right).

Research Questions

1. How do sucralose concentrations in San Marcos and Comal springs compare to concentrations reported in other U.S. groundwaters, surface water bodies, and wastewater treatment plant effluents?
2. How do the sucralose concentrations at Comal and San Marcos springs compare with each other and what is their relationship to total springflow?

Methods

Water samples were collected from Hotel Spring in the San Marcos system and Spring Run 3 in the Comal system monthly from January 2021- June 2025. Sample bottles were submerged directly into flowing springs and not under the influence of surface water runoff. Field duplicates and field blanks were also collected in a similar manner. All sample bottles were kept chilled during transport and frozen until shipment conducted quarterly.

Sucralose analysis was conducted by commercial laboratory SGS AXYS Analytical Services following their standard procedures MLA-116. Briefly, aqueous unfiltered samples were spiked with surrogate standard and acidified. Analysis of the sample extract was performed on an ultra-high performance liquid chromatograph coupled to a triple quadrupole mass spectrometer. Sample blanks did not indicate any issues with lab or field contamination.

Spring discharge data were collected from USGS gaging stations using the R package dataRetrieval (2.7.20).

Discussion

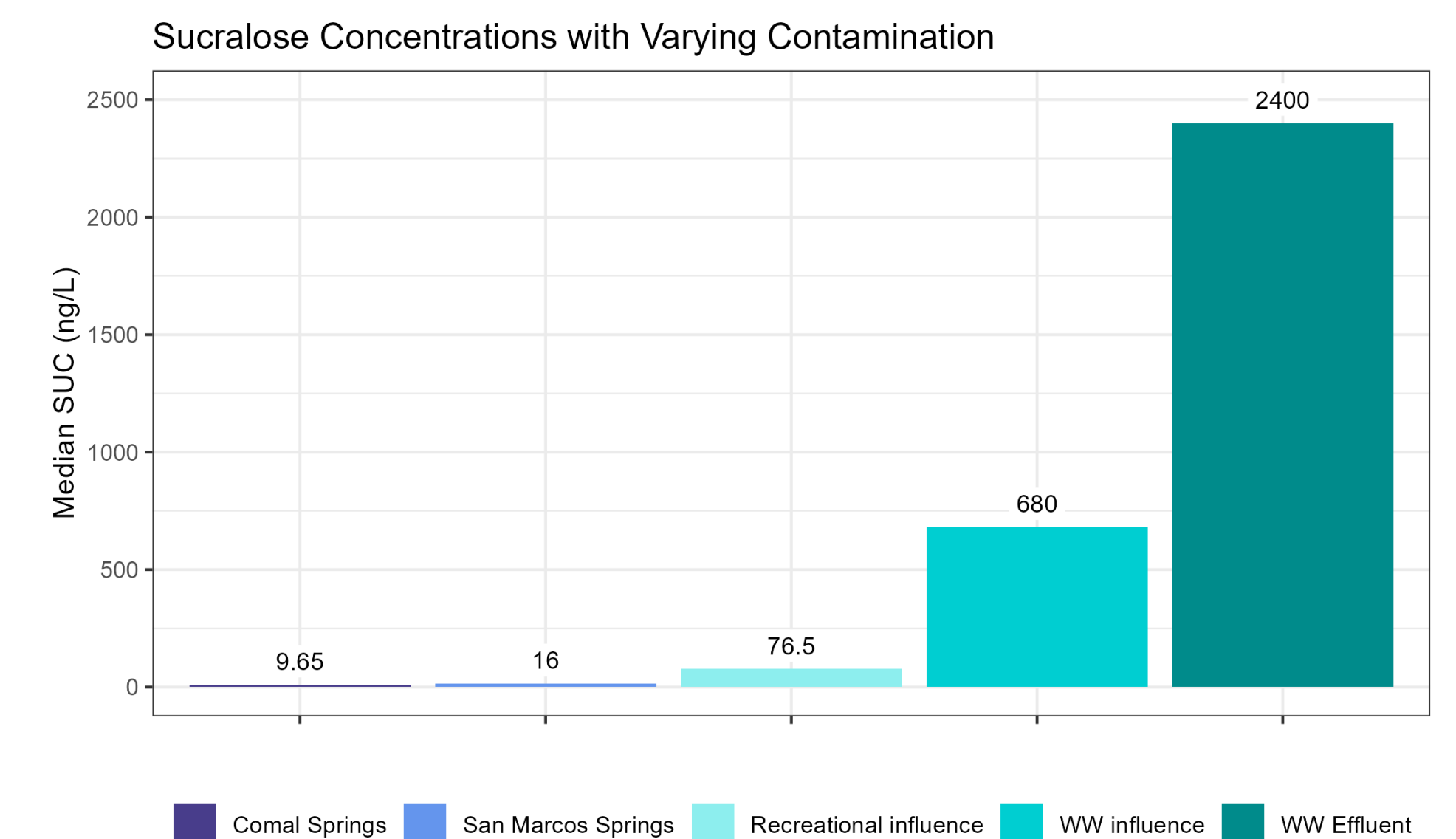


Figure 3: Sucralose concentrations in the Edwards Aquifer springs groundwater compared to average U.S. surface waters with recreational influence³, surface waters with wastewater influence⁴, and wastewater effluent⁵

Sucralose concentrations in the Edwards aquifer springwater are low compared to U.S. averages for wastewater effluent, groundwater and surface waters which suggests minimal wastewater contributions. Additionally, the Edwards has lower sucralose concentrations than waters with purely recreational influence (with no known wastewater inputs) as documented by Mawhinney et al (2011), further suggesting that the aquifer has minimal anthropogenic contaminant inputs.

Generally, the San Marcos spring system has greater sucralose concentrations (18.6±14.6ng/L) than the Comal system (10.5±5.5ng/L) indicating a larger amount of wastewater influence in this springshed. This difference is reduced but still present when a notable outlier (103ng/L) from the San Marcos system is excluded (16.6±6.3ng/L). Although sucralose concentrations generally do not vary with springflow (as shown in Figure 2), the variance may be influenced by the Comal systems' higher average springflow (184 cfs) compared to the San Marcos system (121 cfs).

Correlations between spring discharge and sucralose concentration at each location were analyzed both with non-detects removed and non-detects set as half the reporting detection limit. San Marcos results indicated a weak positive relationship. No correlation was present in Comal. In all cases, the p-values indicated no significant correlation (p-values 0.15-0.81).

References

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